Hawking Cooling: Per-Searcher Step-Size Control from Black-Hole Evaporation, and a Controlled Test of Whether It Helps

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October 2026

Abstract

Annealing-type optimisers usually lower a temperature by a schedule that is global, prescribed in advance, and at most reset by an external trigger. We describe a step-size rule in which each searcher in a population carries its own temperature, derived from a mass that grows with the improvements it finds and shrinks by Hawking evaporation while it stagnates. Because a black hole's temperature is inversely proportional to its mass, the rule has negative heat capacity: during consecutive failures a searcher heats up and its mass loss accelerates until it is replaced by fresh searchers. Cooling, reheating and restarts are therefore driven by each searcher's history, through a discrete mass update and a fixed burst threshold, rather than by a global clock, and the patience a searcher is granted grows with the improvement it has accumulated. We derive continuous approximations to the temperature trajectory and survival time in closed form, and test the rule in a controlled 2×2 factorial design (reheating on or off, earned or fixed patience) against a fixed step size and a per-searcher 1/5 success rule, with shared proposal, selection and burst mechanics. On nine problems (six synthetic functions in ten dimensions and three data-based tasks), the Hawking rule beats a fixed step size on six problems, but its distinctive ingredients provide no consistent benefit, and the 1/5 control is significantly better on eight of nine problems under unadjusted per-problem tests. We report this negative result as the main finding, and identify a likely cause. Because mass grows only in proportion to normalised improvement, each searcher's temperature has a floor that does not shrink with the budget: in our runs the Hawking step never fell below 0.016 (in unit-cube units, about one-twelfth of its initial value), whereas the 1/5 step reached 10^{-3} or less on every problem. Results for a larger algorithm built around the rule are given in an appendix.

1 Introduction

Simulated annealing [5, 15] controls the balance between exploration and exploitation with a temperature that is lowered over time. In most variants the temperature is a single global quantity following a prescribed schedule. Re-annealing [12] rescales temperatures using search sensitivities, while iterated local search and restart evolution strategies [2, 16] perturb or reinitialise the search state. Parallel tempering in its classical replica-exchange form [11] gives each replica its own temperature, but those temperatures are fixed ladder rungs, not quantities each replica earns.

This paper studies a different construction, taken from black-hole thermodynamics. A Schwarzschild black hole of mass M radiates at the Hawking temperature $T \propto 1/M$ [9, 10] and loses mass at a rate $\dot{M} \propto -1/M^2$ [22]. Because its temperature rises as it loses energy, it has negative heat capacity, a property shared with self-gravitating systems [17, 31], and its evaporation accelerates as mass decreases. We use this idealised mass-loss law as a search analogy; the algorithm replaces the low-mass endpoint by an explicit restart.

We map this onto a population of searchers. Each searcher is a "hole" whose mass grows when it finds improvements (accretion) and decays while it does not (evaporation). Its mutation step length is its Hawking temperature $T = k/M$. The resulting rule has three properties that, taken together, we have not found in the step-size adaptation literature:

1. **Per-searcher, unscheduled cooling.** Every searcher has its own temperature, driven by its own successes and failures. There is no global schedule or iteration-dependent decay.

2. **Reheating with runaway.** Failure heats a searcher, and the heating accelerates; a stuck searcher drifts towards more and more exploratory steps before it is restarted.

3. **Earned patience.** The number of consecutive failures a searcher survives grows approximately as a power of its post-success mass, so heavier searchers receive more time.

Metaphor-based metaheuristics are rightly met with scepticism [4, 27]: a new vocabulary often disguises a known mechanism. We therefore state the rule as plain update equations (Section 3), derive its behaviour (Section 4), and compare it against controls that switch off its distinctive ingredients one at a time within a shared search loop (Sections 5 and 6). The result is negative: in our setting the distinctive ingredients provide no consistent benefit, and a fifty-year-old rule [25] does better. We think this is worth reporting, both to save others the effort and because the failure has an identifiable cause: tying mass to accumulated improvement puts a budget-independent floor under each searcher's temperature (Proposition 4 and Section 7).

2 Related work

Annealing and reheating. Simulated annealing lowers a global temperature on a fixed schedule [5, 15]. Very fast simulated re-annealing [12] periodically rescales per-parameter temperatures using sensitivity estimates. Generalised simulated annealing [32, 34] supplies the visiting and acceptance framework used by SciPy's hybrid `dual_annealing` implementation [33]. In that implementation, re-annealing is triggered when the scheduled temperature reaches a specified fraction of its initial value [30]. These mechanisms differ from a temperature coupled to each searcher's accreting and evaporating mass.

Per-replica temperatures. Replica Monte Carlo [29] is an early multi-temperature approach. In classical replica exchange [11], configurations are exchanged between slots with fixed temperatures. Temperature belongs to the slot rather than being earned or lost by a configuration.

Step-size adaptation in evolution strategies. The 1/5 success rule [25, 26] increases the step length when the measured success frequency exceeds 1/5 and reduces it when it falls below that target. The classical formulation measures success over blocks of trials [3]; our controls use a per-trial multiplicative variant. Self-adaptation and cumulative step-size adaptation, as in CMA-ES [3, 6], provide other mechanisms for adapting mutation scale and shape. The Hawking rule moves the step size in the *opposite* direction: success shrinks it and failure enlarges it.

Escalation and abandonment. Basic variable neighbourhood search [20] advances through a predefined sequence of neighbourhoods after failure and returns to the first after improvement; the neighbourhoods need not be nested. The artificial bee colony algorithm [13] uses a common fixed limit without improvement to trigger food-source abandonment. Iterated local search [16] combines local optimisation with perturbation and acceptance decisions, which can depend on search history. Restart CMA-ES [2] reinitialises the search distribution and increases the population size across restarts. These methods provide broader precedents for escaping stagnation than a single fixed failure limit.

Black-hole metaheuristics. The original Black Hole algorithm [7] treats the best solution as an attractor and replaces candidates that cross its event horizon with random ones. Its variants are reviewed by Abualigah et al. [1]. Nemati and Momeni's fuzzy black-holes algorithm uses "Hawking radiation" for positional mutation with a fitness-dependent mutation probability [21], rather than the evaporating-mass proposal scale studied here. Related physics-inspired methods use gravitational attraction between fitness-weighted masses [24] or black-, white- and wormhole transfer between candidate universes [19]. This comparison concerns the mechanisms in the cited works; it is not a systematic novelty search and does not establish a comprehensive priority claim.

3 Method

We consider minimisation. The search runs in the unit cube $[0, 1]^d$, mapped affinely onto the problem's box, so that step lengths are comparable across problems. Each searcher i holds a position x_i, its objective value f_i, a mass M_i and a starvation counter n_i (iterations since its last success). Let s be the standard

deviation of the objective over the initial population, held fixed throughout the run to make mass gain unit-free; if it is zero, we use $s = 1$. Here "temperature" means mutation scale, not an acceptance temperature: only strictly improving proposals are accepted.

Temperature and moves. The temperature of searcher i is

$$T_i = k/M_i, \tag{1}$$

and it proposes $x' = \text{reflect}\Big(x_i + (T_i/\sqrt{d})\, z\Big)$ with $z \sim \mathcal{N}(0, I_d)$, where reflect mirrors points back into the cube. Before reflection, the factor $1/\sqrt{d}$ gives $\mathbb{E}\|(T_i/\sqrt{d})z\|^2 = T_i^2$: T_i is the root-mean-square proposal length, not its expected Euclidean length. Reflection changes the realised length.

Accretion. If $f(x') < f_i$, the searcher moves and gains mass in proportion to the improvement:

$$M_i \leftarrow M_i + \min\left(g\,\frac{f_i - f(x')}{s},\ M_i\right), \qquad n_i \leftarrow 0. \tag{2}$$

The cap stops one lucky step from more than doubling the mass. The implementation also caps the resulting mass at $M_{\max} = 10^5$. The position and stored objective value are then updated to x' and $f(x')$.

Evaporation. Otherwise the searcher stays put and evaporates by a discretised $\dot{M} = -\alpha/M^2$, with a time step equal to its starvation counter:

$$n_i \leftarrow n_i + 1, \qquad M_i \leftarrow M_i - \alpha\, n_i/M_i^2. \tag{3}$$

Success causes no evaporation and resets n_i. Weighting a failed update by n_i rather than 1 further accelerates mass loss during a failure streak. This starvation weighting is an algorithmic addition to the Hawking mass-loss law. Section 6 also tests the plain version with time step 1.

Final burst. When $M_i < M_{\min}$, the searcher is replaced by a child at $\text{reflect}(x_i + (k/M_{\min})/\sqrt{d}\, z)$ with mass m_c, and up to $b - 1$ further children are added while the population is below a cap of $2N$. In the implementation, replacement happens immediately: additional children are centred on the first child rather than the old parent. Each child is evaluated and its starvation counter starts at zero; additional children enter the next population sweep. The best evaluated point is retained separately, even if its searcher bursts.

Algorithm 1 gives the complete loop. Table 3 in Appendix A lists the mass, step-size and population parameters. Every objective call, including initialisation and burst children, counts against the budget; the run stops immediately when it is exhausted.

4 Analysis

Consider a searcher whose last success left it with mass M_{s}, and treat the failure count n as continuous.

Proposition 1 (Negative heat capacity)**.** *Under* (1)*, $C = dM/dT = -k/T^2 = -M^2/k < 0$. Mass loss raises the temperature, and mass gain lowers it.*

Proposition 2 (Continuous temperature trajectory and survival time)**.** *Approximate the discrete failure updates by $dM/dn = -\alpha n/M^2$, starting from $M(0) = M_{\mathrm{s}} > M_{\min}$. Then $M(n)^3 = M_{\mathrm{s}}^3 - \frac{3}{2}\alpha n^2$, and*

$$T(n) = k\left(M_{\mathrm{s}}^3 - \tfrac{3}{2}\alpha n^2\right)^{-1/3}, \qquad n^* = \sqrt{\frac{2\,(M_{\mathrm{s}}^3 - M_{\min}^3)}{3\alpha}}. \tag{4}$$

Here n^* is the continuous threshold-crossing time; it scales as $M_{\mathrm{s}}^{3/2}$ when $M_{\mathrm{s}} \gg M_{\min}$. For the unit-step approximation $dM/dn = -\alpha/M^2$, the corresponding time is $n^* = (M_{\mathrm{s}}^3 - M_{\min}^3)/(3\alpha)$, scaling as M_{s}^3.

Proof. Multiplying the first differential equation by $3M^2$ gives $d(M^3)/dn = -3\alpha n$. Integrating and solving $M(n^*) = M_{\min}$ gives the result. The unit-step case instead has $d(M^3)/dn = -3\alpha$. □

Algorithm 1 Hawking cooling for a population of N searchers.

initialise $x_i \sim U[0,1]^d$, $f_i \leftarrow f(x_i)$, $M_i \leftarrow M_0$, $n_i \leftarrow 0$ for $i = 1..N$
$s \leftarrow \mathrm{std}_i\, f_i$ (use 1 if zero); retain the best evaluated point
while budget remains **do**
  for all searchers i present at the start of this sweep **do**
    draw $z \sim \mathcal{N}(0, I_d)$; $T_i \leftarrow k/M_i$; $x' \leftarrow \mathrm{reflect}(x_i + (T_i/\sqrt{d})\, z)$
    $f' \leftarrow f(x')$; update the best evaluated point
    if $f' < f_i$ **then**
      $M_i \leftarrow \min\{M_{\max}, M_i + \min(g\,(f_i - f')/s,\, M_i)\}$
      $x_i \leftarrow x'$; $f_i \leftarrow f'$; $n_i \leftarrow 0$ ▷ accretion: cools
    else
      $n_i \leftarrow n_i + 1$; $\quad M_i \leftarrow M_i - \alpha\, n_i / M_i^2$ ▷ evaporation: heats
    end if
    if $M_i < M_{\min}$ **then**
      evaluate a replacement child at scale $k/M_{\min}$; set its mass to m_c, counter to 0
      evaluate up to $b-1$ extra children around that child, subject to the cap $2N$
      retain the best evaluated point; queue extra children for the next sweep
    end if
  end for
end while
return best evaluated point

These expressions are not exact solutions of the implemented Euler update. For its jth consecutive failure, cubing (3) gives

$$M_j^3 = M_{j-1}^3 - 3\alpha j + \frac{3\alpha^2 j^2}{M_{j-1}^3} - \frac{\alpha^3 j^3}{M_{j-1}^6}. \tag{5}$$

Dropping the higher-order terms yields $M_n^3 \approx M_{\mathrm{s}}^3 - \frac{3}{2}\alpha n(n+1)$. This approximation requires small relative decrements $\alpha j / M_{j-1}^3$ and can fail near the burst threshold. The actual burst occurs at the first integer j for which $M_j < M_{\min}$.

Proposition 3 (Runaway)**.** *In the continuous approximation, $dT/dn = \alpha n\, T^4/k^3$, which increases in both n and T. A failing searcher heats slowly at first and then ever faster, up to the burst temperature $k/M_{\min}$.*

Proof. $dT/dn = -(k/M^2)\, dM/dn = k\alpha n/M^4 = \alpha n\, T^4/k^3$. □

Figure 1 plots (4) for the default parameters. A searcher with $M_{\mathrm{s}} = 1$ reaches the continuous threshold at about 8 failures; one with $M_{\mathrm{s}} = 10$ reaches it at about 258. These are illustrative values for $\alpha = 0.01$, not the tuned value 0.001 used for Hawking in the main comparison. The temperature stays near its post-success value for most of that time and rises steeply only at the end.

A final property limits how cold a searcher can become.

Proposition 4 (Temperature floor)**.** *Let a searcher be created with mass m (either M_0 or m_c) at a point with objective value f_{b}, and let f^* be the minimum of the objective over the box. At every iteration of its lifetime,*

$$M_i \le m + g\,\frac{f_{\mathrm{b}} - f^*}{s}, \qquad T_i \ge \frac{k}{m + g\,(f_{\mathrm{b}} - f^*)/s}. \tag{6}$$

Proof. Evaporation and the ceiling $M_{\max}$ only reduce mass, and a success from f_i to f' adds at most $g\,(f_i - f')/s$. Because the stored value f_i only decreases, these increments telescope to at most $g\,(f_{\mathrm{b}} - f_i)/s \le g\,(f_{\mathrm{b}} - f^*)/s$. □

The bound depends on the starting point and on the initial spread s, but not on the budget or on the number of successes. For a random start, $(f_{\mathrm{b}} - f^*)/s$ is typically a few units, so with $g = 2$ the temperature cannot fall much more than an order of magnitude below k/M_0. The no-reheating variants of Section 5 share the accretion law and hence the floor. Section 7 reports the scales reached in practice.

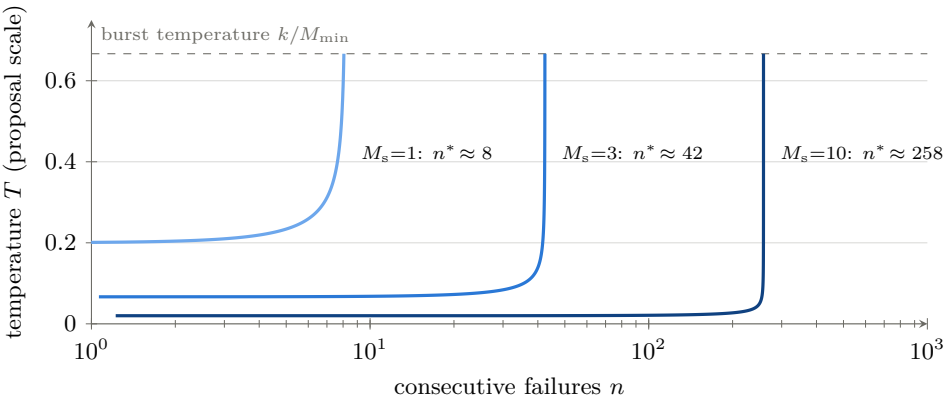


Figure 1: Temperature of a stagnating searcher after its last success at mass M_s, from (4) with $k = 0.2$, $\alpha = 0.01$, $M_{\min} = 0.3$. Each curve ends at the continuous burst threshold. Heavier searchers stay cold longer; the threshold time scales as $M_s^{3/2}$ for masses well above $M_{\min}$.

5 Experimental setup

Problems. Nine minimisation problems: sphere, Rosenbrock, Rastrigin, Ackley, Griewank and Schwefel in $d = 10$; maximum elevation on a 30 m digital elevation model of the Jacksboro Fault, Tennessee ($d = 2$, error in metres); and k-means clustering of the Iris ($d = 12$) and Wine ($d = 39$) data sets, with reference values taken from the best of 50 k-means++ restarts with random seed zero. For clustering, features are standardised to zero mean and unit variance, and the objective is the mean squared distance to the nearest of three centres. These reference values are empirical, not certified global optima. We report $e = \max(f - f^{\text{ref}}, 0)$, using known optima for the synthetic functions and the highest grid elevation for terrain. Clipping prevents negative errors from roundoff or a better-than-reference solution; zero does not certify optimality.

Controls. All variants share the loop of Algorithm 1: population $N = 20$, Gaussian proposal standard deviation $T/\sqrt{d}$ per coordinate, reflection, and a three-child burst at temperature $k/M_{\min}$ with a population cap of 40. Mergers and all other modules of the larger algorithm (Appendix B) are switched off, so the variants differ only in how a searcher sets its step length and when it gives up:

- **Hawking**: the rule of Section 3 (reheating, earned patience).
- **Hawking, $\Delta t = 1$**: evaporation with a unit time step (survival $\propto M_s^3$).
- **No reheating, earned patience**: the same mass-based burst criterion as Hawking, but the step length stays at k/M_s, the value set by the last success.
- **Reheating, fixed patience**: Hawking temperature, saturating at $k/M_{\min}$, but the burst comes after L consecutive failures.
- **No reheating, fixed patience**: step length k/M_s; burst after L failures.
- 1/5 **rule, fixed patience**: per-searcher step σ, multiplied by 1.5 on success (capped at 1) and by $1.5^{-1/4}$ on failure. This is a per-trial controller inspired by the 1/5 success rule [3, 25]; burst after L failures, analogous to the fixed abandonment limit in artificial bee colony [13].
- 1/5 **rule, earned patience**: the 1/5 step rule, with burst timing from the evaporating mass as in Hawking.
- **Fixed step**: step k/M_0 for initial searchers and k/m_c for burst children, constant within each lifetime; burst after L failures.

Hawking, no reheating with earned patience, reheating with fixed patience, and no reheating with fixed patience form the 2×2 design. Each is tuned separately. The contrasts compare complete tuned variants, rather than isolating a change at a common parameter setting or estimating a formal factorial effect. In the no-reheating, fixed-patience cell, mass is not evaporated; it only increases on success and resets at a burst. Thus this cell also changes the mass from which subsequent success-driven cooling proceeds.

Table 1: Median final error over 15 seeds (10 000 evaluations). Lower is better; **bold** marks the best median per problem and the best mean rank. The last column counts problems on which Hawking is significantly better / no difference is detected / significantly worse (Mann–Whitney U, $p < 0.05$).

variant	tuned	sph	rosen	rastr	ackley	griew	schwef	terrain	iris	wine	rank	W/T/L
Hawking	α=.001	7.6e-3	23.2	37.8	17.8	0.878	1170	4.17	1.9e-3	2.98	5.56	–
Hawking, $\Delta t = 1$	α=.01	8.5e-3	21.6	42.8	18.0	0.880	1170	2.68	1.6e-3	2.91	5.11	0/8/1
no reheat, earned	α=.003	9.3e-3	23.5	36.2	16.9	0.871	1100	3.42	2.0e-3	2.97	5.00	1/8/0
reheat, fixed L	L=100	1.1e-2	37.2	33.9	13.5	0.874	1060	3.45	2.2e-3	2.82	4.78	1/7/1
no reheat, fixed L	L=100	1.2e-2	19.4	38.6	16.4	0.899	1070	3.76	2.0e-3	2.99	5.67	0/8/1
1/5 rule, fixed L	L=100	**3.4e-7**	**3.03**	**27.9**	**1.65**	**0.040**	**1050**	**8.8e-9**	**4.2e-6**	**2.08**	1.56	0/1/8
1/5 rule, earned	α=.001	**3.4e-7**	3.34	**27.9**	2.04	**0.040**	**1050**	4.4e-7	4.3e-6	**2.08**	**1.44**	0/1/8
fixed step	L=300	0.479	134	50.6	7.20	2.64	1580	5.95	3.5e-2	2.32	6.89	6/1/2

Protocol. Each run has a budget of 10 000 objective evaluations counted by a shared wrapper. Every variant gets one tuned parameter, chosen by mean rank on screening seeds 100–104 from a four-value grid: $L \in \{8, 30, 100, 300\}$ for fixed patience, and $\alpha \in \{0.001, 0.003, 0.01, 0.03\}$ for earned patience. The reported comparison uses 15 different seeds (0–14). Differences are tested per problem with a two-sided Mann–Whitney U test [18] at $p < 0.05$, without correction for multiple comparisons. We treat these as exploratory comparisons, not tests of equivalence. The test compares distributions rather than medians alone; win/loss labels use the direction of the observed median difference. Mean rank is the average, across problems, of ranks of the unrounded median errors; exact ties are ordered by the fixed variant order in the analysis script. Appendix A describes the implementation and reproducibility materials.

6 Results

Table 1 gives the results. We draw four conclusions.

The adaptive rule improves on the fixed-step control. Hawking has significantly lower error on six problems and higher error on two (Ackley and Wine). This supports the combined adaptive rule relative to this particular fixed-step baseline; it does not isolate the mass–temperature coupling from its restart criterion.

Reheating provides no consistent benefit. With earned patience, Hawking improves significantly on the no-reheating variant only on sphere. With fixed patience, reheating is significantly worse on Rosenbrock and better on Ackley. The mean ranks of the four factorial cells range from 4.78 to 5.67. These mixed results do not establish that reheating has no effect.

Earned patience provides no consistent benefit. With reheating, earned patience is significantly better on sphere and worse on Ackley than the fixed limit. Without reheating, it is significantly worse on Rosenbrock, with no detected difference on the other eight problems. With the 1/5 step rule, no difference between earned and fixed patience is detected on any problem. The unit-step Hawking variant is significantly better than the starvation-clock variant only on terrain. A non-significant comparison does not establish performance equivalence.

The 1/5 controls perform best. Both 1/5 variants have significantly lower error than Hawking on eight of nine problems, with no detected difference on Schwefel. Several differences span orders of magnitude (sphere: 3×10^{-7} versus 8×10^{-3}; terrain: 10^{-8} versus 4 m for fixed patience). These conclusions use the unadjusted tests specified above.

7 Discussion

A likely cause of the performance gap: the temperature floor. Proposition 4 bounds each searcher's temperature from below by a quantity that does not depend on the budget, and Table 2 shows that the bound is reached in practice. On every problem, neither Hawking nor its no-reheating counterpart ever proposed a step smaller than 0.016, about one-twelfth of the initial scale. The 1/5 controller changes its scale by a fixed factor per trial, whatever the size of the improvement, and reached 10^{-3} or less on all

Table 2: Smallest proposal scale (unit-cube RMS step) used by any searcher in a run, median over seeds 0–14 at the selected settings of Table 1. The initial scale is $k/M_0 = 0.2$.

variant	sph	rosen	rastr	ackley	griew	schwef	terrain	iris	wine
Hawking	0.016	0.023	0.023	0.017	0.016	0.017	0.023	0.018	0.017
no reheat, earned	0.016	0.023	0.023	0.017	0.016	0.017	0.022	0.018	0.017
1/5 rule, fixed L	2.2e-5	1.2e-4	3.5e-6	4.6e-7	8.8e-6	9.8e-9	7.3e-16	3.3e-5	9.3e-4

nine problems. Near the minimum of a smooth function, sustained progress requires a step that shrinks in proportion to the distance to the optimum [3, 25]. Once that distance falls well below the floor, success becomes rare, and the searcher evaporates and bursts instead of refining. This is consistent with where the gaps in Table 1 are largest: on sphere, terrain and Iris, the 1/5 controls reach errors between two and eight orders of magnitude lower. On Rastrigin, Schwefel and Wine, where final precision matters less than which basin is found, the medians differ by less than a factor of 1.5. Because the no-reheating variants share the floor, it also explains why switching reheating off does not help. The diagnostic shows that the floor binds. It does not prove that the floor causes the gap, which would require an intervention that removes it.

Feedback direction. A second difference is the sign of the feedback. Each Hawking success adds mass and reduces the next proposal scale, whereas the 1/5 controller increases its scale on success and reduces it on failure. On model problems such as the sphere, progress-rate analysis shows that this keeps the success probability near a useful target [3, 25]. Under the Hawking rule, both directions are self-reinforcing: smaller steps tend to succeed more often and so cool further, while larger steps fail more often and so heat further. The floor caps the first effect and the burst ends the second. We have not separated the contributions of the floor and the feedback direction. Terrain and clustering objectives are only piecewise smooth, so smooth-model theory does not directly cover them.

What survives. The equations do produce the intended qualitative behaviour: unscheduled per-searcher cooling and reheating, and patience that grows with accumulated improvement (Figure 1). Mass links proposal scale and restart patience, although the implemented rule still requires both a starvation counter and a burst threshold. We found no performance benefit from that behaviour here. A version that keeps the evaporation clock only as a restart criterion, and lets a 1/5 rule set the step ("1/5 rule, earned"), has no detected performance difference from the fixed-patience control and needs no tuned abandonment limit L, though it needs a tuned α instead.

Limitations. We tested one dimension for the synthetic problems ($d = 10$), one budget, nine problems and 15 seeds. Grids had four values per variant, and the screening seeds used the same problems as the evaluation. Significance tests were not corrected for multiplicity, and lack of significance is not evidence of equivalence. Shared seed labels do not make the variants follow matched trajectories; the reported Mann–Whitney tests compare their run distributions without using pairing. The temperature-floor diagnostic (Table 2) measures the scales reached; it does not test an intervention. The larger algorithm in Appendix B was developed on the same nine problems. The literature comparison in Section 2 is not a systematic review.

8 Conclusion

We formalised a step-size rule in which each searcher's temperature is the Hawking temperature of an accreting and evaporating mass, derived continuous approximations to its trajectory and $M_s^{3/2}$ survival scaling, and tested its distinctive ingredients in a controlled design. The combined rule outperforms the fixed-step control on six problems, but its reheating and earned patience showed no consistent benefit, and the per-trial 1/5 controller was significantly better on eight of nine problems under exploratory, unadjusted tests. The analogy produces the intended dynamics, but tying mass to accumulated improvement puts a floor under the temperature that limits fine convergence. Removing that floor, for instance with multiplicative accretion, is the natural next test. Such a rule moves towards an inverted success-rule controller, however, and any advantage over the 1/5 rule would have to be shown rather than assumed. Using mass only to time restarts looks less promising: our 1/5 control with earned patience already does this and shows no detected advantage over tuned fixed patience.

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A Implementation details and reproducibility

A.1 Scope of the implementation

The experiments use *Hawking search*, a Python software project containing objective functions, a shared evaluation-budget wrapper, the controlled variants, an extended population optimiser, and baseline implementations. The associated project address is `https://github.com/benzae1/hawkingsearch`. The mathematical method and experimental protocol are specified here so that the paper can be read without navigating the source tree. The companion software documentation provides the installation steps, commands, and mapping from experiments to saved data.

There are three distinct configurations. The main study uses the core rule with *no mergers* and the parameters in Table 3. The *minimal implementation* additionally merges overlapping searchers and is included only in the exploratory comparison in Appendix B. The *extended optimiser* adds population interactions and local refinement, uses separately tuned settings, and is not a controlled test of the cooling rule. The term "Hawking" in Table 1 refers only to the first configuration.

A.2 Core settings and control conventions

Table 3: Core-rule settings for the controlled experiment. α is tuned for earned-patience variants; fixed-patience variants tune L. These settings differ from those of the extended optimiser in Appendix B.

symbol	meaning	value
N	initial population (cap $2N$)	20
M_0	initial mass	1.0
k	temperature constant, $T = k/M$	0.2
g	accretion gain	2.0
α	evaporation constant	0.01 (default)
$M_{\min}$	burst threshold	0.3
m_c	mass of burst children	0.6
b	maximum children per burst	3
$M_{\max}$	numerical mass ceiling	10^5

For controls that retain an evaporating mass, its numerical ceiling is the same as in the core rule. The no-reheating controls hold the proposal scale at k/M_{s} between successes, where M_{s} is refreshed from the current mass after each successful update. The reheating, fixed-patience control uses $\alpha = 0.01$ and clips mass below at $M_{\min}$; its restart is triggered only by L. The 1/5 controls initialise their scale at k/M_0, reset it to k/m_c for burst children, cap it at 1 after success, and impose no lower scale bound. The fixed-step control likewise resets burst children to k/m_c: its scale is constant over a searcher's lifetime, rather than identical for initial searchers and their descendants. All controls share the burst procedure and evaluation accounting described in Section 3.

Away from the upper cap, a success probability p gives expected log-scale change $\log(1.5)\,[p-(1-p)/4]$, which is zero at $p = 1/5$. This explains the target of the chosen per-trial multipliers; the values and cap are experimental settings, not constants prescribed by the original 1/5 rule.

A.3 Data and numerical reproducibility

The controlled study stores one record per variant, parameter setting, problem and seed, including screening and evaluation runs. Its summary uses only seeds 0–14 at each variant's selected setting. Before running the study, the script checks equality of the controlled Hawking implementation with the minimal implementation when mergers are enabled, on Rastrigin with seed 3 and a budget of 2 000 evaluations. This is a regression check on one case, not a proof of implementation equivalence for all inputs. The temperature-floor diagnostic in Table 2 reruns the selected Hawking, no-reheating (earned) and 1/5 (fixed) settings on seeds 0–14 and records the smallest proposal scale used by any searcher in each run.

The terrain data are the Jacksboro Fault elevation sample bundled with Matplotlib; Iris and Wine are the bundled scikit-learn data sets. No network data download is needed for these tasks. Numerical results can depend on library versions and floating-point arithmetic, particularly near machine precision and when a run can terminate in different local basins. A zero in a result table means zero clipped error against the stated reference, not a certified global optimum.

B Exploratory results for the extended optimiser

B.1 Purpose and additional mechanisms

This appendix asks whether a larger optimiser built around the mass rule can be competitive once population interactions and local search are added. It provides context for the software project, not evidence for the two distinctive ingredients tested in the main paper. We describe the additional mechanisms in optimisation terms; the physics names are implementation labels rather than claims of new optimisation principles.

- **Population attraction and diversification.** Searchers drift towards heavier, better members, in the spirit of gravitational search [24], and towards the mass-weighted population centre. Light searchers can make Cauchy-distributed jumps; newly spawned searchers favour regions away from recently evaluated points.
- **Merging and separation.** Overlapping searchers can merge. The algorithm chooses to retain 95% of their combined mass and imposes $M^2_{\text{remnant}} \geq M_a^2 + M_b^2$, motivated by the non-decrease of horizon area [8]. Together, this chosen retention factor and the imposed inequality exclude mass ratios above about 18:1; the factor and cutoff are not universal physical predictions. A diversity-dependent displacement separates nearby searchers.
- **Directional proposals.** A searcher's successful points provide a covariance model for later proposals, related to covariance-based adaptation [6]. Further transformations rotate or stretch proposals near heavier searchers ("spin" and "tidal" terms).
- **Local search and budget allocation.** The leading searcher performs a three-point parabolic line search along learned directions, with an adaptive probe scale. This is called the "Penrose" module, after the energy-extraction analogy of Penrose [23]; that work does not prescribe the numerical line search. During the final 20% of the evaluation budget, only the leader moves.

The extended settings include $N = 10$, a population cap of 10, $k = 0.15$ and $\alpha = 0.001$, whereas the main controls start with $N = 20$ and can grow to 40. Other settings are recorded with the software. Consequently, comparing the extended and core configurations does not isolate the contribution of any single module.

B.2 Comparison with baseline implementations

Table 4 uses the same nine problems, seeds 0–14 and budget of 10 000 evaluations as the main study. Eleven extended-optimiser parameters were selected by random search over 40 settings on seeds 200–205; the best five settings were checked on seeds 100–119. The problems were reused during development, tuning and reporting, so performance estimates may be optimistic for new problems. The baseline configurations did not receive a matched tuning budget. These are comparisons with the supplied implementations, not a ranking of the methods under equally extensive tuning.

Differential evolution and dual annealing use SciPy [28, 33, 34]; dual annealing includes its local search [30], while differential evolution disables polishing. The other baselines are compact implementations: random search, simulated annealing, a $(1 + 1)$ evolution strategy, particle swarm optimisation [14], and CMA-ES with increasing-population restarts [2, 6]. All objective calls, including local-search probes, share the same evaluation limit.

Table 4: Exploratory comparison of the extended optimiser and standard-method implementations (15 seeds, 10 000 evaluations). Errors and mean ranks are defined in Section 5; bold marks the best median per problem and the best mean rank. The "minimal" row enables mergers and differs from the controlled Hawking variant in Table 1.

algorithm	sph	rosen	rastr	ackley	griew	schwef	terrain	iris	wine	rank
dual annealing	3.0e-13	**1.2e-10**	**2.8e-14**	1.8e-8	0.025	**1.3e-4**	8.9e-6	5.9e-11	**4.3e-10**	**2.44**
PSO	9.5e-17	4.12	7.96	1.4e-7	0.057	711	**0**	0.012	1.71	3.22
Hawking, extended	**7.2e-23**	0.299	8.95	2.5e-11	0.103	1620	2.5e-4	1.7e-15	0.664	3.56
CMA-ES	2.0e-13	2.7e-8	6.96	**5.4e-12**	**1.8e-13**	1850	4.3e-11	0.012	5.73	3.89
(1+1)-ES	4.6e-16	1.58	33.8	4.42	0.057	1090	1.1e-6	**1.6e-15**	1.73	4.22
DE	2.0e-5	7.26	30.3	0.052	0.750	712	1.5e-10	2.6e-4	5.22	5.11
Hawking, minimal	1.7e-5	35.5	37.4	16.4	0.245	1030	0.313	4.0e-3	2.90	6.11
simulated annealing	0.053	5320	43.8	0.105	1.11	1460	8.72	0.104	5.96	7.56
random search	12.8	11700	69.0	16.9	45.1	1840	29.5	0.695	10.1	8.89

The extended optimiser has the third-lowest mean rank among these nine implementations. Against dual annealing it has one significant win, three comparisons with no detected difference, and five significant losses under the saved analysis. For this exploratory benchmark and the ablation below, that analysis first treats arrays satisfying NumPy's default `allclose` criterion as ties, then applies unadjusted two-sided Mann–Whitney tests to the remaining comparisons. The controlled study in the main text does not use this numerical-tie precheck. The appendix's significance counts should therefore be read under that additional convention, especially for very small errors.

B.3 What the module ablation supports

Table 5 reports selected leave-one-module-out contrasts using 15 seeds and the same budget. They describe effects within the extended configuration, with all other modules retained; interactions prevent attributing its performance to an independent sum of module contributions.

Table 5: Selected ablations of the extended optimiser. Counts are problems on which the default has significantly lower error / no detected difference / significantly higher error than the ablated version, with the numerical-tie convention described above. Example medians are default → ablated.

removed mechanism	W/T/L	example final error
parabolic line search (Penrose)	6/3/0	sphere: $7.2 \times 10^{-23} \rightarrow 3.5 \times 10^{-5}$
mass-weighted centre attraction	2/6/1	Rastrigin: $8.95 \rightarrow 37.8$
Cauchy jumps	0/9/0	—
space-filling spawning	0/9/0	—

Removing the line search worsens performance significantly on sphere, Rosenbrock, Ackley, Griewank, terrain and Iris. Removing centre attraction worsens Rastrigin and Ackley but improves Schwefel. No significant difference is detected when removing Cauchy jumps or space-filling spawning. These results support the importance of local search in this configuration; they do not establish that every other module is ineffective, or that cooling contributes nothing to the combined optimiser.

B.4 A separate endgame-refinement experiment

A later experiment adds finite-difference BFGS refinement of the leader, called "ringdown", only in the final 20% of the budget. On fresh seeds 500–514, with the same problems and budget, this *endgame-only* variant reaches median errors of 4.86×10^{-15} on Rosenbrock and zero clipped error on Wine. Against dual annealing it has three significant wins (Rosenbrock, Iris, Wine), three comparisons with no detected difference, and three losses, using the appendix's numerical-tie precheck. These counts describe mixed outcomes, not statistical equivalence. They refer specifically to endgame-only refinement; a version that permits refinement earlier has different results. The additional refinement does not change the main study's conclusion about the isolated cooling rule.